

Anionic Hybrid Copolymerization of Sulfur with Acrylate: Strategy for Synthesis of High-Performance Sulfur-Based Polymers

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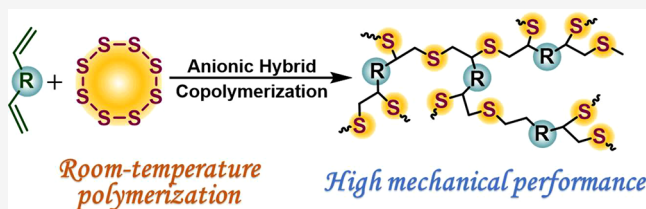


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ABSTRACT: Copolymerization of elemental sulfur (S_8) with vinyl monomers to develop new polymer materials is significant. Here, for the first time, we report the anionic hybrid copolymerization of S_8 with acrylate at 25 °C, yielding a copolymer with short polysulfide segments; i.e., each of them consists of only one to four sulfur atoms. The formation of a longer polysulfide segment would be ceaselessly disrupted by carbon anions through the chain-transfer reaction. The copolymer of S_8 with diacrylate was cross-linked and exhibited excellent mechanical properties, with an ultimate tensile strength as high as 10.7 MPa and a breaking strain of 22%. Furthermore, the introduction of tertiary amide groups to the copolymer enabled it not only to be reprocessed via press molding at room temperature but also to exhibit self-healing properties without external intervention. This study provides a facile strategy to synthesize high-performance sulfur-based copolymers under mild conditions.



INTRODUCTION

Elemental sulfur (S_8) is the third-most abundant element in fossil fuels. Although it has certain applications in rubber vulcanization and additives, it is still excessive. The petrochemical industry alone generates more than 70 million tons of S_8 as a byproduct every year.^{1,2} Developing sulfur-containing polymer materials provides a route to utilize S_8 .^{3–6} For this purpose, S_8 was employed to directly react with diamines, aryl halides, or vinyl comonomers.^{7–10} Also, it was used in multicomponent polymerization with diacids and diamines, or diisocyanides and diamines, etc.^{11–14}

Inverse vulcanization has emerged as the most attractive strategy for synthesizing sulfur-containing polymers in recent years.¹⁰ Unlike classical vulcanization, inverse vulcanization employs a large amount of S_8 relative to the vinyl comonomers to generate sulfur radicals so that they are added to the carbon–carbon double bonds. This process generally occurs above 159 °C, at which point S_8 is converted into polymerizable sulfur. Accordingly, comonomers with low boiling points cannot be employed, and the toxic hydrogen sulfide is likely to be generated.^{15–17} Recently, Quan and co-workers reported photoinduced inverse vulcanization at ambient temperature, which does not generate toxic gases.¹⁸ On the other hand, the sulfur-containing polymers from inverse vulcanization do not have high mechanical properties. So far, the highest reported stress of such polymers was only 8.69 MPa.¹⁰ The poor mechanical performance was attributed to the long polysulfide segments in the backbone with many weak S–S bonds.¹⁹ To improve the mechanical properties, the

copolymer is often chemically cross-linked, which decreases the fraction of sulfur in the product.^{20–23}

S_8 can copolymerize with episulfide via anionic ring-opening copolymerization, where the eight-membered unstrained ring of S_8 is disrupted and reversibly added to an active thiolate anion, yielding sulfur-containing copolymers.^{24–27} Because the active species derived from S_8 was similar to those from episulfide, the copolymerization was highly efficient and generated a high-molecular-weight polymer. However, it is difficult to find applications because of the scarcity and high cost of the episulfide monomer.

Hybrid copolymerization makes it possible that cyclic monomers copolymerize with vinyl monomers into one polymer chain.^{28–31} Such copolymerization is generally difficult because the two types of monomers are quite different in their reactivity and polymerization mechanisms. Recently, we found that S_8 can copolymerize with acrylate via anionic hybrid copolymerization at 25 °C, where the thiolate and carbon anion continuously transfer to each other during the polymerization. Particularly, the copolymer of S_8 and diacrylate exhibits good mechanical properties, reprocessing, and room-temperature self-healing. The present study provides a strategy

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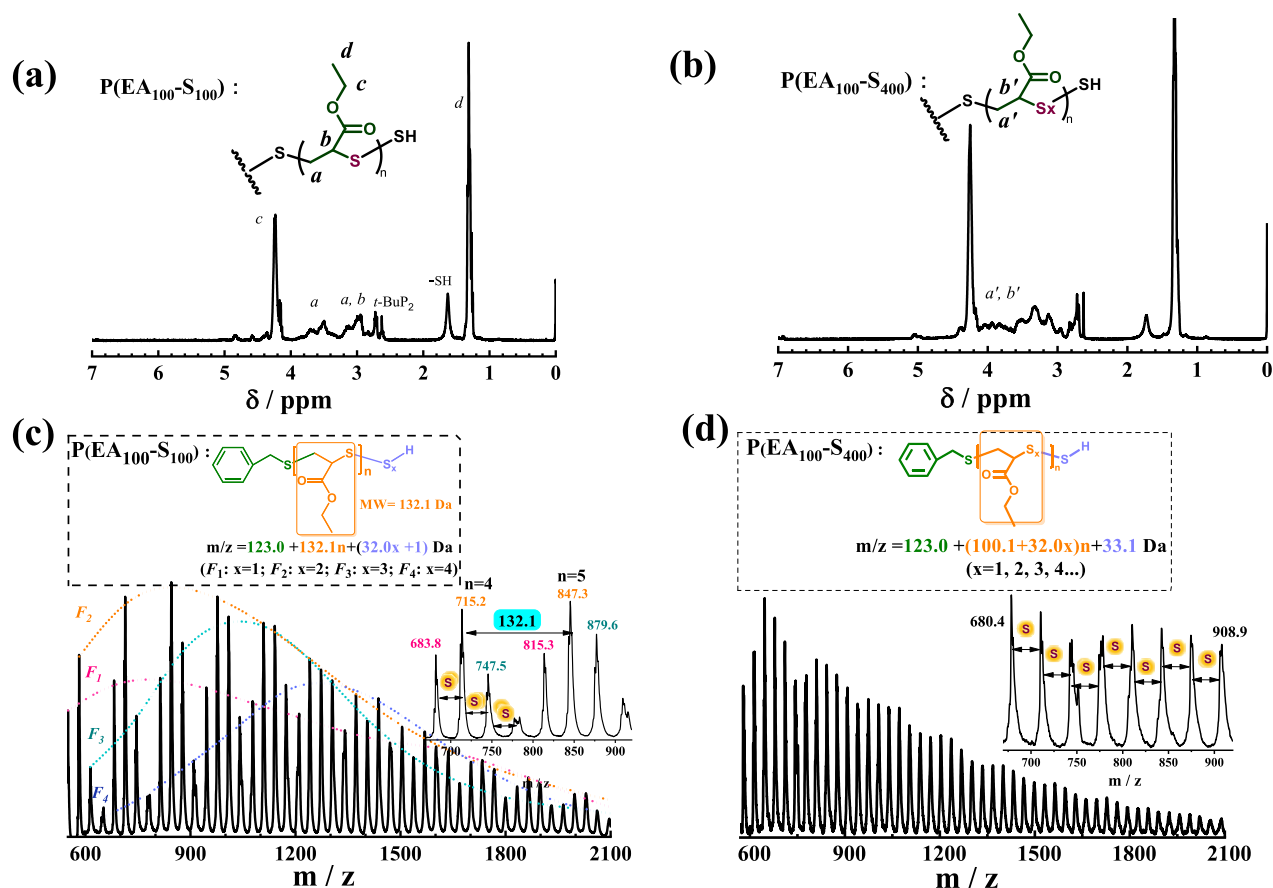


Figure 1. ¹H NMR spectra of (a) P(EA₁₀₀-S₁₀₀) and (b) P(EA₁₀₀-S₄₀₀). MALDI-TOF-MS spectra of (c) P(EA₁₀₀-S₁₀₀) and (d) P(EA₁₀₀-S₄₀₀).

to synthesize high-performance sulfur-based polymers utilizing S₈.

RESULTS AND DISCUSSION

Copolymerization of S₈ with Monoacrylate. The eight-membered ring of S₈ can copolymerize with episulfide via anionic ring-opening copolymerization.^{24,27,32} However, the anionic copolymerization of S₈ and vinyl monomers remains a challenge because of the great difference in their reactivities and polymerization mechanisms.^{17,31} Here, we show that S₈ is able to copolymerize with ethyl acrylate (EA) in the presence of a phosphazene base (*t*-BuP₂)/benzyl mercaptan (BnSH). We explored the possibility for S₈ to copolymerize with other vinyl comonomers, including methyl acrylate (MA), methyl methacrylate (MMA), *N,N*-dimethylaminoethyl acrylate (DMAEA), *N,N*-dimethylaminopropylacrylamide (DMAPEA), acrylonitrile (AN), and styrene (St), by using the initiation system. Actually, only (methyl) acrylate and (methyl) acrylamide were able to copolymerize with sulfur (Figures S1–S4). As discussed below, this is because the carboxyl groups can reduce the electron cloud density of the double bonds so that they are prone to be attacked by anions.

Besides *t*-BuP₂, we also examined other catalysts such as 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU), 1,5,7-triazabicyclo[4.4.0]dec-5-ene (TBD), 7-methyl-1,5,7-triazabicyclo[4.4.0]dec-5-ene (MTBD), and phosphazene base *t*-BuP₁. Together with BnSH, each catalyst system showed high efficiency for the copolymerization (Table S1, Figures S5 and S8), but *t*-BuP₂/BnSH had the highest. In addition, the basicity of the catalyst significantly affected the polymerization rate. For example, the

conversion of EA reached 100% with the molar ratio [EA]:[S]:[BnSH]:[Catalyst] = 100:100:1:1 at 25 °C in 24 h by using *t*-BuP₂ (pK_a = 33.5), while it was only 66% in the case of DBU (pK_a = 21.5). Actually, a small amount of ([S]/[*t*-BuP₂] = 800:1) could effectively catalyze the copolymerization.

Now, we turn to the copolymerization of S₈ with EA in the presence of *t*-BuP₂/BnSH. The color of the reactants quickly turned from dark red to light yellow after a small amount of hydrochloric acid solution was added, which is characteristic of anionic polymerization (Figure S9). Note that a conventional radical initiator such as 2,2'-azobis(2-methylpropionitrile) (AIBN) could not initiate the copolymerization. Furthermore, the introduction of a radical scavenger, (2,2,6,6-tetramethylpiperidin-1-yl)oxyl (TEMPO), to the system could not change the polymerization. All the facts indicate that the polymerization follows an anionic mechanism instead of a radical one.

S₈ and EA were completely consumed only when the molar ratio of [EA]:[S] was in the range of 1:1 to 1:4. Beyond this range, either unreacted EA or S₈ was observed (Table S1). Since EA could not homopolymerize under the same conditions, the length of the polysulfide segment in the copolymer can be calculated from the feed ratio. As the sulfur content increased from 50 to 80 mol%, the segment contained one to four sulfur atoms. This is quite different from the case for the inverse vulcanization. The latter has long polysulfide segments with weak S–S bonds connected by vinyl comonomer units, which would reduce its mechanical property.^{33,34}

Proton nuclear magnetic resonance (¹H NMR) was employed to analyze the structure of the product. P(EA₁₀₀-

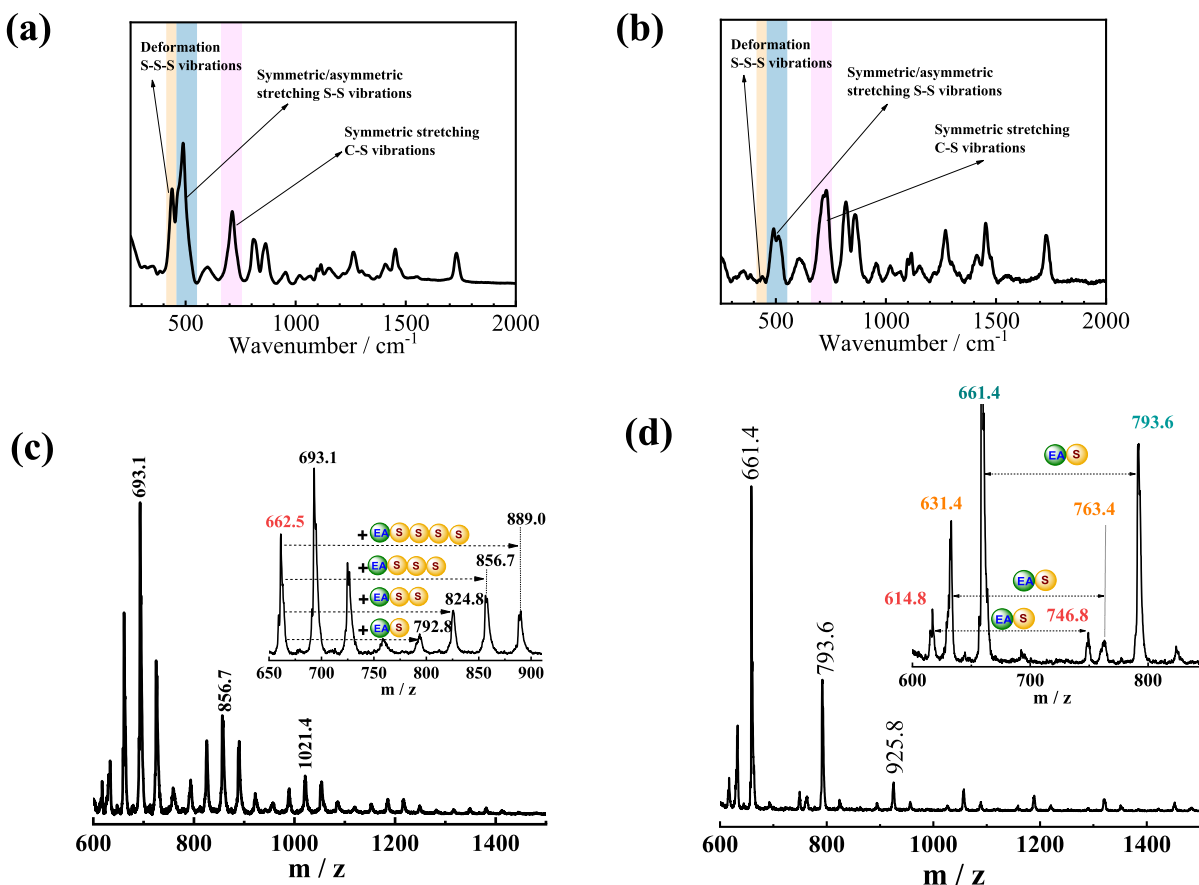


Figure 2. Raman spectra of P(EA₁₀₀-S₁₀₀) at polymerization times of (a) 1 and (b) 24 h. MALDI-TOF-MS of P(EA₁₀₀-S₁₀₀) spectra at polymerization times of (c) 1 and (d) 24 h.

S₁₀₀) and P(EA₁₀₀-S₄₀₀) were prepared using the molar ratios [EA]:[S]:[BnSH]:[*t*-BuP₂] = 100:100:1:1 and 100:400:1:1, respectively. As shown in Figure 1a,b, the peaks at 1.1–1.4 and 4.1–4.3 ppm were assigned to methyl (*d*) and methylene (*c*) protons in the EA unit. For P(EA₁₀₀-S₁₀₀), the peaks assigned to methylene (*a*) and methyne (*b*) protons bonded to a single sulfur atom were observed at 2.8–3.2 and 3.3–3.7 ppm, respectively. For P(EA₁₀₀-S₄₀₀), besides those signals, other peaks at 3.8–4.2 ppm were also observed. These signals were assigned to the methylene (*a'*) and methyne (*b'*) protons bonded to the polysulfide blocks. This could be further confirmed by the thiolysis experiment (Figure S10). In short, NMR measurements indicated that the repeat units of P(EA₁₀₀-S₁₀₀) and P(EA₁₀₀-S₄₀₀) were S–C–C and S_{*x*}–C–C (1 < *x* ≤ 4), respectively.

The products were analyzed using matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI-TOF-MS). Figure 1c shows that the peaks in the LS-MS distribution of P(EA₁₀₀-S₁₀₀) were spaced by 132.1 *m/z*, corresponding to the S–EA unit in the backbone, further indicating that the repeat unit of the copolymer consists of S–EA and the sulfur atoms were alternately distributed in the chain with S–C–C structure. In addition, there were four fractions in P(EA₁₀₀-S₁₀₀). Actually, they were different from each other in sulfur number at the chain end. The peaks in the LS-MS distribution of P(EA₁₀₀-S₄₀₀) were spaced by 32.1 *m/z* instead of 132.1 *m/z* (Figure 1d). This is understandable because the repeat units are composed of polysulfide segments with different sulfur numbers (S_{*x*}–C–C, *x* = 1, 2, 3, 4). This is

consistent with the NMR results. In sum, the polysulfide segment can be readily tuned by varying the feed ratio.

Gel permeation chromatography (GPC) measurements show that P(EA₁₀₀-S₁₀₀) and P(EA₁₀₀-S₄₀₀) had low molecular weights of ~1000 g mol⁻¹ (Figure S11). This is because the large amount of S–S bonds promote the chain-transfer reactions and inhibit the formation of a high-molecular-weight polymer.

Polymerization kinetics was also studied. ¹H NMR measurements show that EA conversion was 50% in 1 h and reached 95% in 24 h, where the molar ratio of [EA]:[S]:[BnSH]:[*t*-BuP₂] was 100:100:1:1 (Figure S12). On the other hand, the length of the polysulfide segment in the copolymer varied with time. Raman spectra show that the length of the polysulfide segment decreased with time (Figure S13). The band for S–S–S vibrations could be clearly observed at 410–500 cm⁻¹ in 1 h (Figure 2a).³² However, such signals almost disappeared when the reaction time reached 24 h (Figure 2b). Furthermore, the band for the C–S vibration at 620–715 cm⁻¹ considerably increased in intensity. The ¹H NMR spectra (Figure S14) show that the signals of methyne and methylene protons bonded to the polysulfide segments at 3.8–4.2 ppm were present in 1 h, but they disappeared in 24 h. This agrees with the MALDI-TOF-MS results in Figure 2c,d.

The copolymerization was also studied by a density functional theory (DFT) calculation. As shown in Figure 3, the Gibbs free energy change (Δ*G*) for the carbon anion reacting with an EA molecule was 48.7 kcal mol⁻¹, much higher than that (9.89 kcal mol⁻¹) for the carbon anion

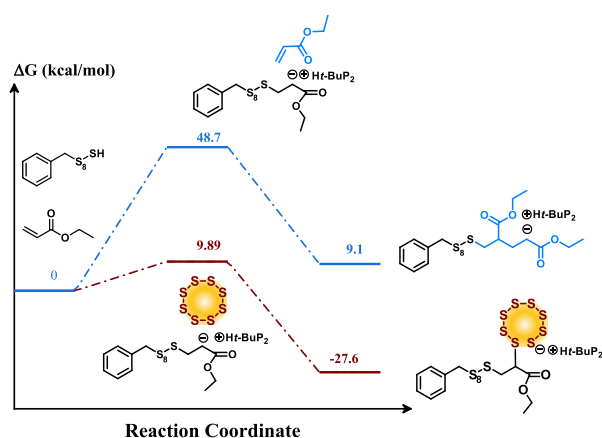


Figure 3. Gibbs free energy profile for the anionic hybrid copolymerization of S_8 and acrylate at 25 °C calculated by using DFT methods.

reacting with one S_8 . Accordingly, the carbon anion was more prone to attack the sulfur atom than EA, forming a new active carbon anion. The alternating process (red line) had an ΔG of $-27.6 \text{ kcal mol}^{-1}$, so it was thermodynamically favorable, which ensures the copolymerization continuing. Besides, EA is more reactive with S_8 than MA because EtO^- has a higher nucleophilicity than MeO^- . This was evidenced by the DFT calculation in Figure S15. Namely, the ΔG for reaction of MA with S_8 is $29.3 \text{ kcal mol}^{-1}$, much higher than that for EA with S_8 .

With the data presented above, the mechanism of anionic hybrid copolymerization of S_8 with acrylate catalyzed by $t\text{-BuP}_2$ at 25 °C is proposed (Figure 4). BnSH is first deprotonated by

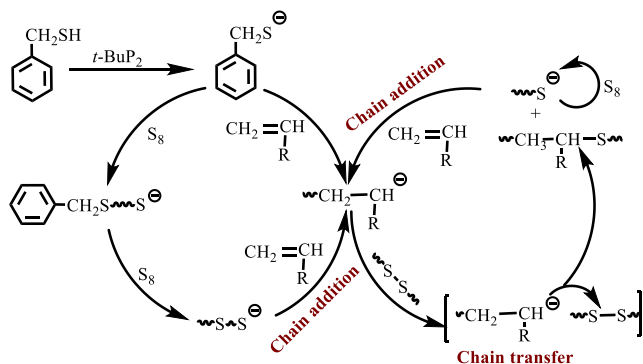


Figure 4. Proposed mechanism for the anionic hybrid copolymerization of S_8 and acrylate at 25 °C.

$t\text{-BuP}_2$ to generate a living thiolate anion that can open the sulfur ring to form a polysulfide anion or react with a double bond to form a carbon anion. The former further reacts with a double bond to generate a carbon anion. Meanwhile, the latter attacks and breaks the weak S–S bond so that the active species transfer from carbon to sulfur to form a polymeric chain with an active thiolate anion. The S–S bonds in the dormant polymer can be further activated via a new transfer reaction. In short, the copolymerization is a hybrid of ring-opening polymerization and stepwise polyaddition, which is regulated by chain transfer. Note that the copolymer synthesized via anionic hybrid copolymerization has short polysulfide segments. This is because longer polysulfide

segments with more weak S–S bonds are more readily broken, as Pyun et al. reported before.³⁵

Copolymerization of Sulfur with Diacrylate. Since monoacrylate comonomers yield only low-molecular-weight polymers, we had to synthesize a cross-linked polymer by using 1,4-butanediol diacrylate (BDDA) to copolymerize with S_8 so that the polymer had good mechanical properties. The copolymerization happened at 25 °C, and the reaction system was completely solidified in 24 h when the feed ratio of $[\text{BDDA}]:[S_8]:[t\text{-BuP}_2]:[\text{BnSH}]$ was 60:40:0.5:0.5 in a solvent mixture of tetrahydrofuran (THF) and carbon disulfide (CS_2) (1:1, v/v). CS_2 did not react with the sulfur anion (Figure S16). The product could not dissolve in a solvent such as N,N -dimethylformamide (DMF), methylene dichloride (MDC), THF, or CS_2 , indicating its cross-linking (Figure 5). Note that

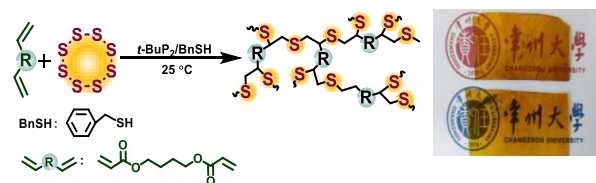


Figure 5. Schematic illustration of the copolymerization of S_8 and diacrylate catalyzed by $t\text{-BuP}_2$ at 25 °C.

neither BDDA nor S_8 can homopolymerize under the same conditions (Figure S17), so the product should be a cross-linked copolymer. Besides, no H_2S was released during the polymerization because of the low reaction temperature. In contrast, the inverse vulcanization polymerization of BDDA and S_8 conducted at 180 °C yielded only cross-linked BDDA homopolymer with unreacted S_8 (Figure S18), accompanied by release of H_2S .

The copolymer was characterized by Raman spectra (Figures 6a and S19). The bands at $410\text{--}500 \text{ cm}^{-1}$ were assigned to the deformation of S–S–S bonds, and the bands at $510\text{--}540 \text{ cm}^{-1}$ were assigned to the symmetric/asymmetric stretching S–S vibrations. Particularly, the band at $620\text{--}715 \text{ cm}^{-1}$ assigned to the symmetric stretching C–S vibration was observed, indicating the formation of a sulfur-containing copolymer.^{27,32} The Fourier transform infrared (FTIR) spectrum (Figure 6b) shows bands at 483 and 670 cm^{-1} caused by S–S stretching and C–S stretching, respectively, further indicating the formation of the copolymers.^{10,36,37} X-ray diffraction (XRD) measurements (Figure 6c) show that S_8 had characteristic Bragg reflections at 23.2° , 25.9° , and 27.9° , corresponding to the (222), (026), and (040) planes, respectively. In contrast, the sharp crystalline sulfur peaks disappeared in $\text{P}(\text{BDDA}_{16}\text{S}_{84})$, indicating that the product was an amorphous copolymer without unreacted sulfur.

X-ray photoemission spectroscopy (XPS) in the presence of C 1s, O 1s, and S 2s and 2p provided more evidence (Figure 6d). The high-resolution C 1s XPS spectrum (Figure 6e) shows a series of characteristic peaks located at 284.8, 285.3, 286.4, and 289.3 eV, corresponding to species of carbon–carbon (C–C) and carbon–heteroatom (C–S, C–O, and C=O) bonds, respectively.^{16,38} The characteristic peaks at 163.6, 163.9, 164.8, and 165.1 eV in the high-resolution S 2p XPS spectra (Figure 6f) were assigned to the S 2p $1/2$ and S 2p $3/2$ of S–S species and their doublet of C–S species.³⁸ Integration of the corresponding peaks gave the ratio of S-to-C–S bonds to be 1.37, close to the theoretical value (1.16)

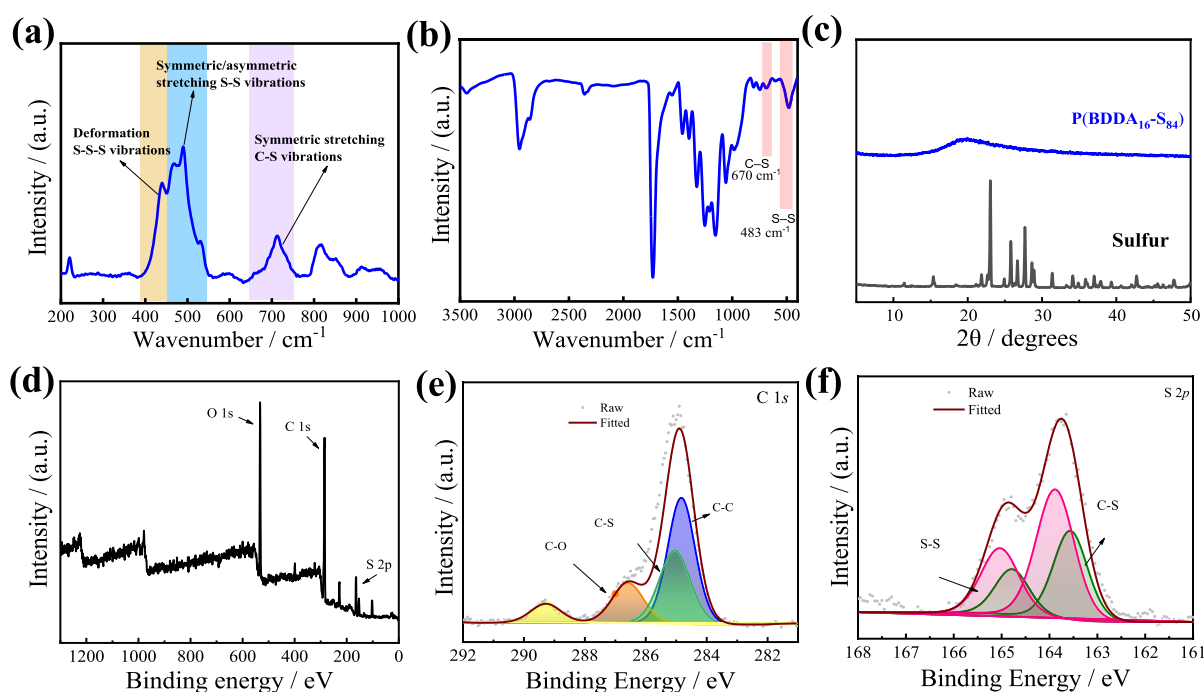


Figure 6. (a) Raman spectrum of P(BDDA₁₆-S₈₄). (b) FTIR spectrum of P(BDDA₁₆-S₈₄). (c) XRD patterns of P(BDDA₁₆-S₈₄) (top) and S₈ (bottom). (d) XPS survey scan of P(BDDA₁₆-S₈₄). (e) C 1s XPS and its deconvoluted peaks featured for C–O, C–S, and C–C. (f) S 2p XPS spectrum and its deconvoluted peaks of S–S and C–S.

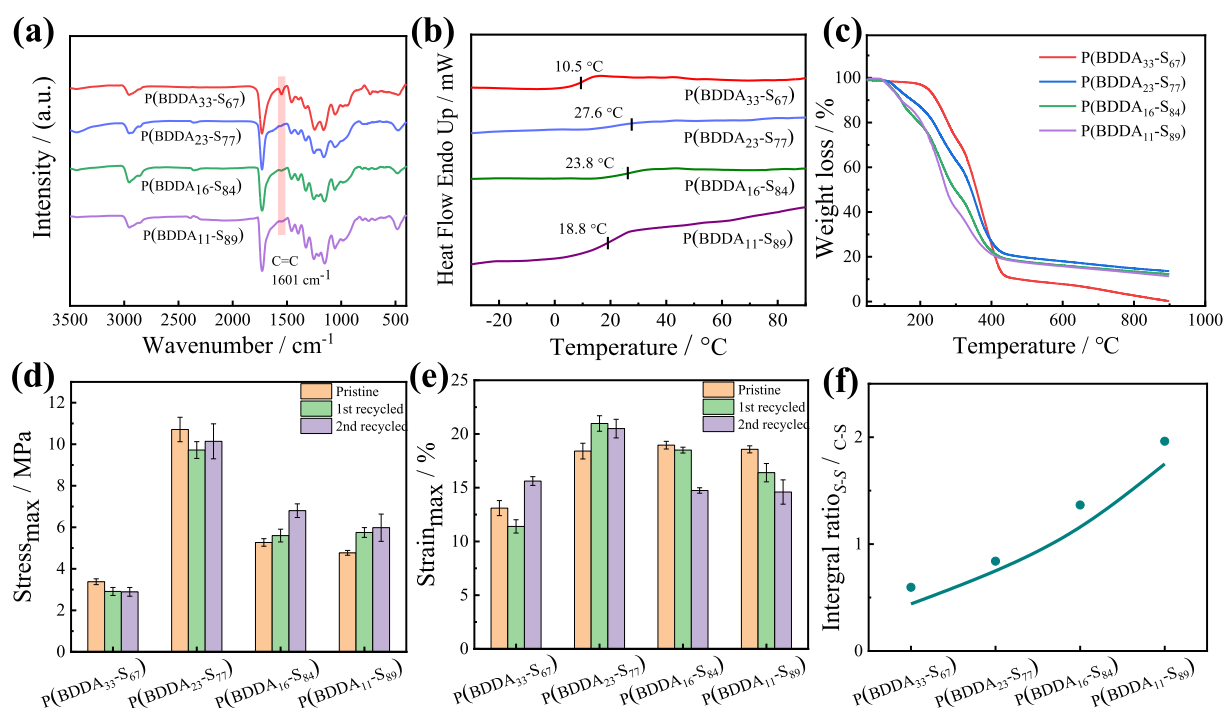


Figure 7. (a) FTIR spectra, (b) DSC thermograms, (c) thermogravimetry (TGA) curves, (d) ultimate tensile stress at break, (e) maximum strain at break, and (f) molar ratio of the S–S to C–S bonds calculated from XPS of P(BDDA_x-S_y) copolymers with different sulfur contents.

calculated from the monomer feeds by assuming each C=C bond forms two C–S bonds. All the facts indicate the successful synthesis of the sulfur-based cross-linking copolymer.

Thermal and Mechanical Properties. A series of copolymers were synthesized by varying the sulfur content from 67 to 89 mol% in the feed; i.e., the feed ratio of [C=C]:[S] was changed from 1:1 to 1:4 (Table S2). The product was

termed P(BDDA_x-S_y), where *x* and *y* represent the loadings of BDDA and S in molar percent, respectively. FTIR spectra (Figure 7a) show that P(BDDA₃₃-S₆₇) had a band at 1601 cm^{−1}, assigned to unreacted C=C bonds. Namely, it had a low cross-linking density and poor mechanical performance. The band was absent in P(BDDA₂₃-S₇₇), P(BDDA₁₆-S₈₄), and P(BDDA₁₁-S₈₉), indicating that the vinyl monomers were completely polymerized.

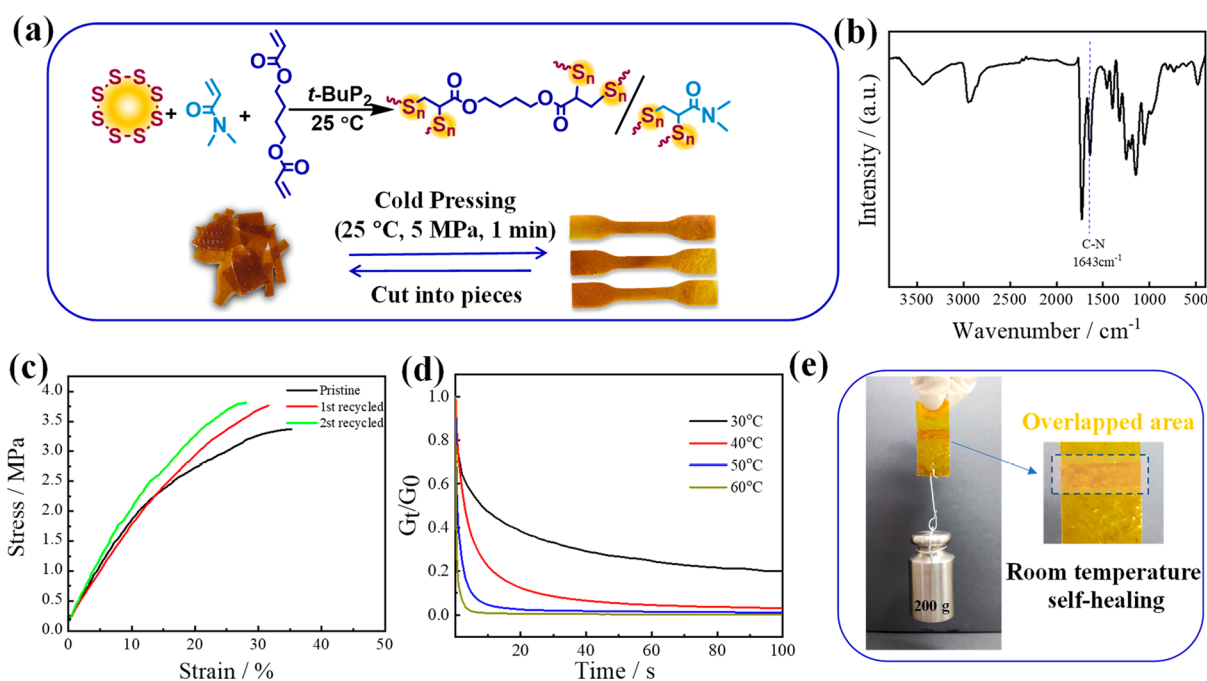


Figure 8. (a) Schematic illustration of the synthesis and reprocessing of copolymers of S_8 , BDDA, and DMA. (b) FTIR spectrum of $P(BDDA_{10}\text{-}DMA_2\text{-}S_{88})$. (c) Typical strain–stress curves of reprocessed $P(BDDA_{10}\text{-}DMA_2\text{-}S_{88})$. (d) Stress relaxation of $P(BDDA_{10}\text{-}DMA_2\text{-}S_{88})$ at 30, 40, 50, and 60 °C. (e) Photograph of the self-healing sample.

Figure 7b shows the differential scanning calorimetry (DSC) curves of the copolymers. As the sulfur content increased from 67 to 77 mol%, the glass transition temperature (T_g) increased from 10.5 to 27.6 °C. This is understandable because the segment motion becomes more restricted as the cross-linking density increases. Further increasing the sulfur content caused T_g to decrease to 18.8 °C. This is due to the excessive sulfur that can reduce the cross-linking density.

Figures 7c and S20 show the thermogravimetric analysis (TGA) curves and derivative thermogravimetric analysis (DTGA) curves. The copolymer with a sulfur content of 67 mol% began to lose weight at 210 °C. However, it exhibited considerable weight loss starting at ~110 °C when the sulfur content increased up to 77 mol%. Namely, the thermal decomposition temperature (T_c) dropped as the sulfur content increased. This is because $P(BDDA_{33}\text{-}S_{67})$ is mainly composed of S–C–C units, which are much more stable than the $S_x\text{-C-C}$ units in $P(BDDA_{23}\text{-}S_{77})$. Further increasing the sulfur content led to weaker S–S bonds, so $P(BDDA_{16}\text{-}S_{84})$ and $P(BDDA_{11}\text{-}S_{89})$ showed an even lower decomposition temperature.

Figure 7d,e shows the ultimate tensile stress and maximum strain of the copolymers. $P(BDDA_{23}\text{-}S_{77})$ had an ultimate tensile strength as high as 10.7 MPa with a breaking strain at 22%. To our knowledge, this is the highest value for copolymers directly prepared from sulfur and vinyl comonomers. The excellent mechanical property can be attributed to the short polysulfide segments and the high cross-linking density. As shown in Figure 7f, the S–S to C–S ratio (0.8) of $P(BDDA_{23}\text{-}S_{77})$ calculated based on the XPS spectra in Figures S21–23 was close to the theoretical value (0.75) calculated from the feed ratio, indicating that the copolymer had C–S–S–C and C–S–C sequences. Note that $P(BDDA_{23}\text{-}S_{77})$ s synthesized with different amounts of catalyst, e.g., $[S]/[t\text{-BuP}_2] = 100:1$ and $400:1$, did not make a difference in the

strain–stress curves, indicating that the remaining catalyst had very limited effect on its mechanical properties (Figure S24).

It is interesting that the copolymer exhibited reprocessability. It was repeatedly press-molded under 10 MPa at 100 °C in 5 min, and most of the mechanical properties were able to hold after two “press-molded–cut into pieces” cycles (Figure 7d,e). FTIR and XRD spectra (Figures S25 and S26) show no crystallinity of free sulfur during the reprocessing. Unlike sulfur-containing polymers from inverse vulcanization, the copolymer here did not change its structure in the processing. Clearly, the dynamic nature of the S–S bonds should be responsible for this.

Anionic hybrid copolymerization also enables ready chemical modification of the polymer. Chalker et al. reported that some amide-based solvents such as DMF and dimethylacetamide (DMAC) could reduce the energy required to cleave S–S bonds so that the two surfaces of the sulfur polymers could be chemically joined at room temperature.⁹ To further improve the reprocessability of the copolymer, tertiary amide groups were introduced by copolymerization with *N,N*-dimethylacrylamide (DMA) (Figure 8a,b). $P(BDDA_{10}\text{-}DMA_2\text{-}S_{88})$ copolymer showed a breaking strain of 30% (Figure 8c), higher than that (22%) of the corresponding copolymer without DMA. This is because the presence of DMA increases the length of the polymer segment between the two cross-linking points. As a result, the former showed more rapid reprocessability than the latter. It was repeatedly press-molded under 5 MPa in 1 min at 25 °C or under 5 MPa within seconds at 40 °C. After two processing cycles, the mechanical properties almost did not change (Figure 8c).

DSC measurements show that $P(BDDA_{10}\text{-}DMA_2\text{-}S_{88})$ had a T_g of 16.7 °C (Figure S27). The normalized stress–relaxation curve (Figure 8d) shows that it could completely relax and flow at a temperature above T_g . The obtained relaxation time (τ^*) ranged from 1 min at 30 °C to 0.4 s at 60 °C. So far, it exhibited the most rapid relaxation at low temperature among

the sulfur-based polymers. On the other hand, the activation energy for the dynamic bond reaction is 117 kJ mol^{-1} (Figure S28), higher than those for the copolymers synthesized via inverse vulcanization.³⁹ This is because the ratio of C–S (strong bond) to S–S (weak bond) in the former is much higher than that in the latter.

Particularly, P(BDDA₁₀-DMA₂-S₈₈) exhibited self-healing without external intervention at room temperature. As shown in Figure 8e and Movie S1, the P(BDDA₁₀-DMA₂-S₈₈) sample was first cut in half, and then the two halves were partially overlapped without heating or applying force. They joined together 24 h later, and the healed sample was able to load a 200 g weight. Clearly, the cold press-molding and room temperature self-healing properties are attributed to the enhanced dynamic sulfur exchange reactions due to the tertiary amide in DMA units.⁹

CONCLUSION

Elemental sulfur (S₈) can copolymerize with acrylate to form a sulfur-based copolymer at 25 °C via anionic hybrid copolymerization. The copolymer consists of a short polysulfide segment with one to four sulfur atoms. The formation of a longer polysulfide segments would be ceaselessly disrupted by carbon anions through a chain-transfer reaction. The hybrid copolymerization of S₈ with diacrylate leads to a copolymer with short polysulfide segments and high cross-linking density, so it exhibits high mechanical properties. The introduction of tertiary amide groups enables the cross-linked copolymer not only to be repeatedly cold-press-molded but also to self-heal at room temperature without external intervention. The present study provides a facile strategy for the synthesis of high-performance or functional sulfur-containing copolymers.

EXPERIMENTAL SECTION

Materials. Sulfur (S₈, colloidal powder, reagent grade, Aladdin), 1,4-butanediol diacrylate (BDDA, 80%, Aladdin), MA (98%, Aladdin), EA (98%, Aladdin), *N,N*-dimethylaminopropylacrylamide (DMAPAA, >99%, Aladdin), methyl methacrylate (MMA, 98%, Aladdin), *N,N*-dimethylaminoethyl acrylate (DMAEA, >99%, Aladdin), and styrene (St, 80%, Aladdin) were distilled prior to use. *N,N*-Dimethylacrylamide (DMA, >99%, Aladdin), tetrahydrofuran (THF, >99%, Aladdin), benzyl mercaptan (BnSH, >99%, Aladdin), *t*-BuP₂ (2 mol L⁻¹, solution in THF, Aladdin), 1,8-diazabicyclo[5.4.0]undec-7-ene (DBU, >99%, Aladdin), 1,5,7-triazabicyclo[4.4.0]dec-5-ene (97%, Aladdin), 7-methyl-1,5,7-triazabicyclo[4.4.0]dec-5-ene (MTBD, 98%, Aladdin), *t*-BuP₁ (P₁, >97%, Aladdin), and anhydrous carbon disulfide (CS₂, >99%, Sigma) were all used as received. CS₂ is a colorless, toxic liquid and is used as a solvent.

Measurements. The NMR spectra were recorded by using a Bruker Avance DRX (400 MHz) spectrometer. FTIR spectroscopy was conducted using a Vertex 70 Fourier transform infrared spectrometer. Raman spectroscopy was performed using a Thermo Fisher Scientific DXRTM 2 Raman microscope with 532 nm laser excitation. DSC was performed using a Q6000 DSC instrument at a heating/cooling rate of 10 °C min⁻¹ at temperatures ranging from -50 to 100 °C. TGA was conducted using a TA Instruments Q500 under nitrogen from 30 to 900 °C at a heating rate of 20 °C min⁻¹ under a nitrogen atmosphere. XRD was conducted using a PANalytical X'Pert PRO diffractometer with Cu K α radiation. A tensile test was performed using a WDT-10 electronic universal testing tensile machine at 25 °C, following ASTM Standard D412 with a crosshead rate of 10 mm min⁻¹. The average values with error bars of all mechanical data were obtained after 3–6 tests for every sample. XPS was performed using a Thermo ESCALAB with a monochromatic Al K α X-ray source having an energy of 1486.6 eV.

Pb(OAc)₂ test paper was used to detect H₂S under different reaction conditions. The molecular weights were measured at 35 °C in THF by using a Waters 1515 GPC instrument equipped with a Waters 2414 differential refractive index detector. HPLC-grade THF was used as the eluent at a flow rate of 1.0 mL min⁻¹. MALDI-TOF-MS mass spectra were recorded on a Bruker Autoflex Speed MALDI-TOF/TOF.

General Procedure for the Synthesis of Copolymers. A typical procedure for the synthesis of P(BDDA_{*x*}-S_{*y*}) is as follows. S₈ (20 mmol, 5.12 g), BDDA (20 mmol, 3.96 g), CS₂ (4 mL), and THF (4 mL) were charged into a 250 mL round-bottom flask equipped with a magnetic stirrer in a glovebox. Then, the reaction vessel was sealed and removed from the glovebox, and *t*-BuP₂ (50 μ L, 0.2 mmol, 1 equiv) was introduced with BnSH (25 μ L, 0.2 mmol, 1 equiv) to start the polymerization. After reacting at 25 °C for 24 h, the resulting cross-linked copolymer was placed in 100 mL of a CS₂:HF (1:1 v/v) solution for 2 h to remove unreacted monomers. Finally, the copolymer was dried at 60 °C in a vacuum oven.

Mechanical Properties. Stress–Relaxation. Stress–relaxation tests of polymers with different sulfur contents were conducted using an Anton Paar MCR 301 in the linear viscoelastic region at different temperatures (30, 40, 50, and 60 °C) at a constant strain of 2%. The activation energy (*E*_a) of stress relaxation was calculated according to the Maxwell model between the relaxation time and temperature. Under the designed temperature, the stress–relaxation behavior (*G*_{*t*}/*G*₀)–time (*t*) dependence was recorded. *E*_a was calculated by fitting the Arrhenius equation:

$$\ln \tau = \ln \tau_0 + E_a/RT$$

where τ is the relaxation time when the value of *G*_{*t*}/*G*₀ reaches *e*⁻¹, *R* is the gas constant (8.314 J K⁻¹ mol), and *T* is the thermodynamic temperature.

Reprocessing. The reprocessing of P(BDDA_{*x*}-S_{*y*}) was performed by using a hot-pressing machine (Yan Zheng Instrument Co., Ltd., China). The sample was chopped into small pieces and hot-pressed at 80–100 °C for 10 min under a constant pressure of 10 MPa. The sample was cooled to room temperature and released from the press.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.3c04746>.

Additional ¹H NMR spectra, FTIR spectra, XRD patterns, XPS spectra, DSC curve, GPC curve, and Arrhenius plots (PDF)

Movie S1, showing self-healing of P(BDDA₁₀-DMA₂-S₈₈) at room temperature (MP4)

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Notes

The authors declare no competing financial interest.

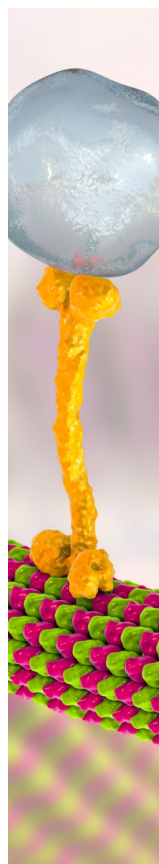
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